

Finite-horizon Weil coercivity and shifted-zeta contraction

A full-output parity certificate through total horizon $9/5$

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Abstract

We study the central Weil form associated with the causal finite-horizon transfer of Suzuki’s shifted zeta ratio. A computer-assisted calculation proves $Q_{0,L} \succeq 10^{-26}I$ for every total horizon $0 < L \leq 9/5$. The outer horizon lies beyond $\log 6$: the generator contains the prime-power delays 2, 3, 4, 5, while the transfer includes the first mixed-prime composition. The proof combines an analytic lower bound on the infinite Legendre tail with full-output Gram matrices retaining the exact reflection-parity coupling. Both resulting 64×64 Schur inequalities are verified by ball arithmetic, with every finite construction operation enclosed and an explicit analytic profile remainder subtracted. An even-in-shift generator identity gives $\|V_{\omega,L}\| \leq \exp(-5 \times 10^{-27}\omega)$ for $0 < \omega \leq 3 \times 10^{-14}$ and $L \leq 9/5$, together with quantitative reflection-positivity and cumulative-storage bounds. The argument does not use zeta zeros or an assumed positive spectral measure. It establishes a finite-depth result, not a zero-free region or an all-depth positivity theorem.

Keywords. Weil positivity; Riemann zeta function; causal convolution; Hankel operators; Schur complement; validated computation.

Review status. The analytic argument and executable certificate are supplied for review. The finite calculation and replay have passed; independent human verification and proof-assistant formalization have not been completed. The numerical result is isolated in Theorem 6.1.

1 Introduction and main result

Weil’s positivity criterion expresses the Riemann hypothesis as positivity of an explicit quadratic form on all compactly supported test functions. Fixing a bounded support produces a separate, finite-horizon question: is the restricted form coercive, and can an explicit positive lower bound be proved without information about zeta zeros? Such a result controls only a bounded part of the criterion, but provides a test of methods intended to extend to larger arithmetic depth.

This paper addresses that question using the causal transfer associated with the shifted quotient

$$\Theta_{\omega}(z) = \frac{\xi(\frac{1}{2} - \omega - iz)}{\xi(\frac{1}{2} + \omega - iz)}, \quad \xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (1.1)$$

Suzuki introduced this family and its arithmetic Hankel kernel [6]. Our statements concern the finite causal convolution defined in Section 2; no global innerness assumption is made.

Throughout the paper, L is the *total length* of the input interval. Its centered version is $(-L/2, L/2)$, and its autocorrelation is supported in $[-L, L]$. The corresponding multiplicative cutoff is $a = e^{L/2}$. This convention is important when comparing arithmetic thresholds or compact-window bounds.

Theorem 1.1 (Finite-horizon coercivity and contraction). *Let*

$$L_* = \frac{9}{5}, \quad m_* = 10^{-26}, \quad h_* = 3 \times 10^{-14}.$$

For every $0 < L \leq L_$, the central form defined in (2.6) satisfies*

$$Q_{0,L}[f] \geq m_* \|f\|_2^2 \quad (f \in \mathcal{D}_{\log,L}). \quad (1.2)$$

For the transfer $V_{\omega,L}$ of Definition 2.1,

$$\|V_{\omega,L}\| \leq e^{-5 \times 10^{-27} \omega} < 1, \quad 0 < \omega \leq h_*. \quad (1.3)$$

Consequently $H_{\omega,L} = V_{\omega,L} R_L$ is self-adjoint and

$$I \pm H_{\omega,L} \succeq (1 - e^{-5 \times 10^{-27} \omega}) I. \quad (1.4)$$

The central estimate is computer-assisted; the remaining implications are analytic.

The total horizon $9/5$ is strictly larger than $\log 6$. At this horizon the logarithmic generator has precisely the delays $\log 2, \log 3, \log 4, \log 5$. The coefficient of $n = 6$ occurs in the transfer as a product of two prime factors, although $\Lambda(6) = 0$. In the full-output calculation the ordered mixed overlaps $(2, 3)$ and $(3, 2)$ are retained explicitly.

The proof has three components. First, differentiating the finite Euler product gives a generator even in ω and a bounded perturbation $Q_{\omega,L} - Q_{0,L} = O_L(\omega^2)$. Second, a fractional-integration estimate gives a lower bound on the infinite Legendre tail. Third, full-output Grams control the coupling between that tail and the retained space, separately in each reflection sector. The finite comparison is a Schur inequality at the claimed positive level m_* ; positivity of a finite compression alone would not suffice.

1.1 Relation to earlier work

Connes and Consani study archimedean Weil positivity through a compressed scaling action and Sonin spaces [1]. Their work motivates the broader question of combining archimedean and arithmetic terms. The present construction uses causal shift differentiation and a polynomial tail estimate.

The recent preprint of Zhu [8], version 2, reports certified compact-window positivity at total input length 1.6, using a frequency-envelope reduction. Its window parameter is a half-width, so its $L = 0.8$ corresponds to total length 1.6 here. Our central form has the same geometric normalization after that conversion. We do not use its numerical certificate. The specific contribution here is the full-output parity reduction, its fractional tail bound, and an independently assembled certificate through total length 1.8, with explicit positive-shift consequences. We make no claim that finite-matrix approaches to Weil positivity are themselves new.

The estimates are finite-depth. In particular, they neither establish global innerness of (1.1) nor prove a new zero-free region. The shift interval is small, and the storage corollary below does not supply an induction in arithmetic depth.

2 Causal transfer and the central form

2.1 Finite-horizon definitions

Let $\mathcal{H}_L = L^2(0, L; \mathbb{C})$, with inner product linear in the second argument, and define

$$R_L f(x) = f(L - x), \quad T_d f(x) = \mathbf{1}_{x>d} f(x - d).$$

In particular, $T_d = 0$ for $d \geq L$ and $T_d T_e = T_{d+e}$. Every adjoint and norm below refers to unweighted $L^2(0, L)$.

Put $\Gamma_\infty(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$. For $0 < \omega \leq 1/2$ the archimedean causal convolution has Laplace transfer

$$K_\omega^\gamma(p) = \frac{\Gamma_\infty(\frac{1}{2} + p - \omega)}{\Gamma_\infty(\frac{1}{2} + p + \omega)}. \quad (2.1)$$

It may be defined on a sufficiently far right Laplace line, followed by restriction to $(0, L)$. For an explicit impulse, set $\alpha = \frac{1}{2} - \omega$, $\beta = \frac{1}{2} + \omega$. Then

$$K_\omega^\gamma(p) = \pi^\omega \frac{(p + \alpha)(p - \beta) \Gamma((p + \alpha)/2)}{(p - \alpha)(p + \beta) \Gamma((p + \beta)/2)}. \quad (2.2)$$

The last ratio is the transform of $2e^{-\alpha t}(1 - e^{-2t})^{\omega-1}/\Gamma(\omega)$. The rational factor has an impulse consisting of a delta mass and smooth exponentials (with the usual limiting interpretation at $\alpha = 0$). Thus the full impulse κ_ω is locally integrable, behaves as $O(t^{\omega-1})$ at the origin, and defines a bounded finite-horizon convolution $V_{\omega,L}^\gamma$.

Definition 2.1. For $\omega > 0$, define

$$b_\omega(n) = n^{\omega-1/2} \prod_{p|n} (1 - p^{-2\omega}), \quad b_\omega(1) = 1, \quad (2.3)$$

$$B_{\omega,L} = \sum_{\log n < L} b_\omega(n) T_{\log n}, \quad V_{\omega,L} = B_{\omega,L} V_{\omega,L}^\gamma. \quad (2.4)$$

Set $V_{0,L} = I$ in the strong-limit sense proved below.

The sum is finite and contains all integers below e^L . This is the logarithmic-coordinate form of Suzuki's arithmetic kernel. The Euler product identity for its Dirichlet series also identifies the full Laplace transfer, on a sufficiently far right line, as $\xi(\frac{1}{2} + p - \omega)/\xi(\frac{1}{2} + p + \omega)$. Real causal convolution gives $V_{\omega,L}^* = R_L V_{\omega,L} R_L$; therefore $H_{\omega,L} = V_{\omega,L} R_L$ is self-adjoint and $\|H_{\omega,L}\| = \|V_{\omega,L}\|$.

2.2 The central Weil form

For $f \in \mathcal{H}_L$, let $f_c(x) = f(x + L/2)$ on $(-L/2, L/2)$ and zero elsewhere. Write

$$\widehat{f}_c(\tau) = \int_{\mathbb{R}} f_c(x) e^{-i\tau x} dx, \quad C(f) = \int_{\mathbb{R}} f_c(x) \cosh(x/2) dx, \quad S(f) = \int_{\mathbb{R}} f_c(x) \sinh(x/2) dx.$$

The logarithmic form domain is

$$\mathcal{D}_{\log,L} = \left\{ f \in \mathcal{H}_L : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{f}_c(\tau)|^2 d\tau < \infty \right\}. \quad (2.5)$$

With $\psi = \Gamma'/\Gamma$, define

$$\begin{aligned} Q_{0,L}[f] = & \frac{1}{2\pi} \int_{\mathbb{R}} (\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi) |\widehat{f}_c(\tau)|^2 d\tau \\ & + 2|C(f)|^2 - 2|S(f)|^2 - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (2.6)$$

Here $\Lambda(p^k) = \log p$ and is zero otherwise. The last scalar is real. The Fourier multiplier differs from $\log(2 + |\tau|)$ by a bounded function, by the digamma asymptotics [4, §5.11]. All remaining terms are bounded. Hence (2.6) is a closed semibounded form on (2.5). It is the central geometric Weil form in this normalization. It commutes with reflection and with complex conjugation. The same notation $Q_{0,L}$ denotes its associated self-adjoint operator where appropriate.

Lemma 2.2 (Polynomial core). *Polynomials restricted to $(0, L)$ form a core for $Q_{0,L}$. If P_N is the orthogonal projection onto the polynomials of degree less than N , polynomial combinations in $(I - P_N)\mathcal{H}_L$ are a form core for the restriction to that subspace.*

Proof. Equip (2.5) with its weighted Fourier norm, equivalent to a shifted form norm. First shrink the centered zero extension by a dilation about the origin so that its support lies in a smaller interval. Dilations tending to the identity converge in this weighted norm: their norms are uniformly bounded, since the logarithmic weights at dilated frequencies are comparable, and strong convergence follows first on smooth Fourier data and then by density. Mollification with support smaller than the resulting boundary gap converges in the weighted norm by dominated convergence. Consequently $C_c^\infty(0, L)$ is a form core.

For $h \in C^1[0, L]$, integration by parts in its zero extension gives

$$|\hat{h}(\tau)| \leq L\|h\|_\infty, \quad |\hat{h}(\tau)| \leq \frac{2\|h\|_\infty + L\|h'\|_\infty}{|\tau|} \quad (\tau \neq 0).$$

These bounds imply continuity from $C^1[0, L]$ to the logarithmic form domain. Approximate the derivative of a smooth function uniformly by a polynomial and integrate, matching one value. This gives polynomial approximation in C^1 , hence in form norm. Endpoint jumps of the zero extension cause no difficulty.

Finally P_N is bounded in form norm: it has finite rank, its range consists of form-domain polynomials, and its coefficient functionals are bounded in L^2 . Apply $I - P_N$ to the polynomial approximants of a vector orthogonal to the retained space. $\square$

3 The prime-delay generator and continuation

3.1 Exact logarithmic differentiation

Write $\ell(p) = \log \Gamma_\infty(\frac{1}{2} + p)$ on a far right half-plane. The archimedean logarithmic generator has symbol

$$\begin{aligned} a_\omega^\gamma(p) &= \ell'(p - \omega) + \ell'(p + \omega) \\ &= \frac{1}{p + \alpha} + \frac{1}{p - \alpha} + \frac{1}{p + \beta} + \frac{1}{p - \beta} - \log \pi \\ &\quad + \frac{1}{2}\psi((p + \alpha)/2) + \frac{1}{2}\psi((p + \beta)/2). \end{aligned} \quad (3.1)$$

It defines a causal logarithmic operator $A_{\omega,L}^\gamma$ on its natural core. Its symmetric form at zero is the gamma part of (2.6). For example, the two central rational terms give the symmetric kernel $2 \cosh((x - y)/2)$, whose form is $2|C|^2 - 2|S|^2$; the digamma term gives the multiplier in (2.6).

Proposition 3.1 (Prime-power generator). *On the causal generator core,*

$$\partial_\omega V_{\omega,L} = -A_{\omega,L} V_{\omega,L}, \quad (3.2)$$

$$A_{\omega,L} = A_{\omega,L}^\gamma - 2 \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) T_{\log n}. \quad (3.3)$$

Thus its symmetric form is

$$Q_{\omega,L} = Q_{\omega,L}^\gamma - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) (T_{\log n} + T_{\log n}^*). \quad (3.4)$$

Proof. The commuting nilpotent delay algebra gives

$$B_{\omega,L} = \prod_{\log p < L} (I - p^{-1/2-\omega} T_{\log p})(I - p^{-1/2+\omega} T_{\log p})^{-1}.$$

Each inverse is a terminating geometric series. Expanding the logarithm of one factor, the coefficient of $T_{k \log p}$ is $(p^{-k/2+k\omega} - p^{-k/2-k\omega})/k$. Its derivative is $2(\log p)p^{-k/2} \cosh(k\omega \log p)$. Differentiating the product in (2.4) and using commutation with causal gamma convolution proves (3.3). Symmetrization gives (3.4). Only the bounded arithmetic factor has been inverted; no bounded inverse of the smoothing transfer is asserted. $\square$

3.2 A bounded quadratic change in the shift

Set

$$r(t) = e^{t/2} - \frac{e^{-5t/2}}{1 - e^{-2t}}, \quad t > 0. \quad (3.5)$$

Lemma 3.2 (Bounded shift difference). *The forms $Q_{\omega,L}$, $0 \leq \omega \leq 1/2$, have the common domain $\mathcal{D}_{\log,L}$. Their gamma difference has symmetric kernel*

$$(\cosh(\omega|x-y|) - 1)r(|x-y|).$$

In particular,

$$\|Q_{\omega,L} - Q_{0,L}\| \leq C_L \omega^2, \quad (3.6)$$

as a bound on the bounded form difference, where

$$\begin{aligned} C_L = & \cosh(L/2) \left(e^{L/2} \frac{L^3}{3} + \frac{L^2}{4} \right) \\ & + \frac{1}{2} \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} (\log n)^2 \cosh(\log n/2) \|T_{\log n} + T_{\log n}^*\|. \end{aligned} \quad (3.7)$$

Proof. In the digamma integral [4, §5.9], form $a_\omega^\gamma - a_0^\gamma$ before separating the divergent terms. Its digamma part has inverse Laplace kernel $-2e^{-t/2}(\cosh(\omega t) - 1)/(1 - e^{-2t})$. The rational terms contribute $4 \cosh(t/2)(\cosh(\omega t) - 1)$. Since $2 \cosh(t/2) - e^{-t/2}/(1 - e^{-2t}) = r(t)$, symmetrization gives the asserted kernel. It is $O(|x-y|)$ at the diagonal and extends continuously in ω through $1/2$. This physical-space identity avoids taking an incomplete pointwise digamma limit at the endpoint.

Now $|r(t)| \leq e^{t/2} + 1/(2t)$ and $0 \leq \cosh(\omega t) - 1 \leq \omega^2 t^2 \cosh(t/2)/2$. The row-integral bound for the symmetric kernel is at most $\omega^2 \cosh(L/2) \int_0^L (t^2 e^{L/2} + t/2) dt$. Bounding each arithmetic difference in (3.4) in the same way gives (3.7). Bounded perturbations preserve the closed-form domain. $\square$

Proposition 3.3 (Central coercivity implies contraction). *If $Q_{0,L} \succeq mI$, then*

$$\|V_{\omega,L}\| \leq \exp(-m\omega + C_L \omega^3/3), \quad 0 < \omega \leq 1/2. \quad (3.8)$$

Proof. We give the regularity argument because the generator is unbounded and $V_{0,L} = I$ is a strong, not a norm, limit. Fix a Laplace line $\Re p = \eta > 1$. Uniform gamma asymptotics, together with absolutely convergent zeta Dirichlet series on this line, give for $0 \leq s \leq 1/2$

$$|K_s(\eta + i\tau)| \leq C(2 + |\tau|)^{-s}.$$

On each $\varepsilon \leq s \leq 1/2$, its first two shift derivatives are bounded by $C_\varepsilon(1 + \log(2 + |\tau|))^j(2 + |\tau|)^{-\varepsilon}$, $j = 1, 2$. Weighted half-line convolution therefore gives norm differentiability on positive-shift intervals and (3.2) after causal restriction. Dominated convergence on the same line gives $V_{s,L}f \rightarrow f$ in $L^2(0, L)$ as $s \downarrow 0$.

For $0 < r < \min(s, 1/2)$ the multiplier maps into H^r locally; multiplication by an interval indicator is bounded on $H^r(\mathbb{R})$ for $r < 1/2$ [5]. After restriction and zero extension, the output lies in $\mathcal{D}_{\log,L}$. The same estimates with a logarithmic factor justify the causal generator action and extension of its symmetric form identity from smooth data. Hence, for every input f , on a compact positive-shift interval,

$$\frac{d}{ds} \|V_{s,L}f\|^2 = -2Q_{s,L}[V_{s,L}f] \leq -2(m - C_L s^2) \|V_{s,L}f\|^2.$$

Gronwall's inequality, first from ε to ω and then with $\varepsilon \downarrow 0$, proves (3.8). No norm derivative at zero or bounded transfer inverse is used. $\square$

4 An analytic lower bound on the infinite Legendre tail

Let $\phi_n(x) = \sqrt{(2n+1)/L} P_n(2x/L - 1)$ and let P_N retain $\phi_0, \dots, \phi_{N-1}$. Define causal fractional integration by

$$I_\nu f(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-y)^{\nu-1} f(y) dy.$$

The following estimate applies to all combinations in the infinite tail.

Lemma 4.1 (Fractional tail estimate). *For $N \geq 1$ and $0 < \nu < 1$,*

$$\|I_\nu(I - P_N)\| \leq (L/2)^\nu \sqrt{\frac{\Gamma(N+1-\nu)}{\Gamma(N+1+\nu)}}. \quad (4.1)$$

Consequently

$$\|I_1(I - P_N)\| \leq \frac{L}{2\sqrt{N(N+1)}}. \quad (4.2)$$

Proof. Termwise beta integration of the terminating Legendre polynomial gives

$$I_\nu P_n(2x/L - 1) = \frac{n!}{\Gamma(n+1+\nu)} x^\nu P_n^{(-\nu,\nu)}(2x/L - 1).$$

The polynomial and orthogonality identities may also be read from [4, §§18.3, 18.5, 18.17]. For a finite tail combination, use $x^{2\nu} \leq (L/2)^{2\nu} x^\nu (L-x)^{-\nu}$ and Jacobi orthogonality. The squared norm becomes bounded by

$$(L/2)^{2\nu} \sum_{n \geq N} |c_n|^2 \frac{\Gamma(n+1-\nu)}{\Gamma(n+1+\nu)}.$$

Successive gamma ratios decrease by the factor $(n+1-\nu)/(n+1+\nu) < 1$. Density and boundedness of I_ν prove (4.1). Letting $\nu \uparrow 1$, using convergence of the fractional kernels in $L^1(0, L)$, gives (4.2). $\square$

The central gamma derivative $U^\gamma = \partial_\omega V_{\omega,L}^\gamma|_0$ has the decomposition

$$U^\gamma = J_L + S_w, \quad J_L = \partial_\nu[(2\pi)^\nu I_\nu]|_0, \quad w(t) = \frac{G_0(t) - 1}{t}, \quad (4.3)$$

where S_w is causal convolution and

$$G_0(t) = \frac{e^{t/2}t}{\sinh t} - 4t \cosh(t/2) = \sum_{j \geq 0} g_j t^j. \quad (4.4)$$

To verify (4.3), the inverse kernel of $-a_0^\gamma$, away from the origin, is $e^{t/2}/\sinh t - 4 \cosh(t/2)$; subtracting the finite-part kernel of J_L leaves w . The constant term agrees with the Laplace asymptotic $-a_0^\gamma(p) = \log(2\pi/p) + O(p^{-1})$. In particular $g_0 = 1$, $g_1 = -7/2$, and $g_2 = -1/24$.

Proposition 4.2 (Central infinite-tail floor). *Suppose $K_L \geq |w(0)| + \int_0^L |w'(t)| dt$. Then, on $(I - P_N)\mathcal{D}_{\log, L}$,*

$$Q_{0,L} \succeq a_{N,L} I, \quad (4.5)$$

where

$$\begin{aligned} a_{N,L} = & H_N - \gamma - \log(\pi L) - \frac{K_L L}{2\sqrt{N(N+1)}} \\ & - \sum_{\log n < L} \frac{\Lambda(n)}{\sqrt{n}} \|T_{\log n} + T_{\log n}^*\|. \end{aligned} \quad (4.6)$$

Proof. Apply (4.1) to a polynomial tail vector, multiply by $(2\pi)^\nu$, square, and differentiate at $\nu = 0$. Both sides have the same value there. The result is $-\operatorname{Re}\langle f, J_L f \rangle \geq (H_N - \gamma - \log(\pi L))\|f\|^2$. The differentiation is legitimate on each polynomial, whose derivative contains only polynomials and logarithmic endpoint terms in L^2 .

Next $S_w = (w(0)I + S_{w'})I_1$. Young's inequality and (4.2) give

$$\|S_w(I - P_N)\| \leq \frac{K_L L}{2\sqrt{N(N+1)}}.$$

Since $Q_0^\gamma = -\operatorname{Re} U^\gamma$ in form sense, this is a lower-bound loss. Bound the remaining symmetric delays individually in norm. Lemma 2.2 extends the resulting polynomial inequality to the complete tail form domain. $\square$

For clarity, the delay norm in (4.6) is not automatically 2. Decompose x into $x = r + kd$, $0 < r < d$. On each fiber, $T_d + T_d^*$ is the adjacency matrix of a finite path. If $q = \lceil L/d \rceil$, its norm is

$$\|T_d + T_d^*\| = 2 \cos \frac{\pi}{q+1} \quad (d < L), \quad (4.7)$$

with threshold equality using the shorter path, since the exceptional fiber has measure zero. In particular the norm is 1 for $d < L \leq 2d$ and $\sqrt{2}$ for $2d < L \leq 3d$. The implementation uses these two exact cases and the safe bound 2 otherwise.

4.1 A uniform analytic profile remainder

Lemma 4.3 (Profile enclosure). *For $0 < L < 3$ and an integer $M \geq 1$, let $\tilde{U}$ retain the terms g_j , $j \leq M$, in (4.3), leaving J_L and the arithmetic delays exact. Then the bounded difference satisfies*

$$\|U - \tilde{U}\| \leq \eta_M, \quad \eta_M = \frac{256(L/3)^{M+1}}{(M+1)(1-L/3)}. \quad (4.8)$$

One may use

$$K_L = \sum_{j=1}^M |g_j| L^{j-1} + \frac{256}{3} \frac{(L/3)^M}{1-L/3} \quad (4.9)$$

in Proposition 4.2.

Proof. On $|z| = 3$, the Euler product for $\sinh z/z$ [4, §4.36] gives $|z/\sinh z| \leq 3/\sin 3 < 24$. Since $e^{3/2} < 5$, (4.4) implies $|G_0(z)| < 5(24 + 12) < 256$. Thus $|g_j| \leq 256 3^{-j}$. The L^1 norm of the omitted convolution kernel is bounded by $\sum_{j>M} |g_j| L^j/j \leq \eta_M$, proving (4.8). For w , the constant term contributes $|g_1|$, while integration of each derivative term gives $|g_j| L^{j-1}$. Summing the omitted geometric series proves (4.9). $\square$

5 Exact reflection-parity coupling and the Schur test

Write $P = P_N$, let P_r be its parity- r part ($r = \pm 1$), and denote the central model form by $\tilde{Q} = -(\tilde{U} + \tilde{U}^*)/2$. The polynomial space lies in the operator domain: the fractional derivative of a polynomial has only square-integrable logarithmic endpoint terms, and all other terms are bounded convolutions or delayed polynomials. Thus $Q_0 P$ and $\tilde{Q} P$ are bounded finite-column operators.

Define the retained matrix and two full-output Grams by

$$q = P\tilde{Q}P, \quad F = (\tilde{U}P)^*(\tilde{U}P), \quad C = (\tilde{U}P)^*R_L(\tilde{U}P). \quad (5.1)$$

In this notation, F and C are finite matrices but the output in their definition has *not* been projected onto $P\mathcal{H}_L$.

Lemma 5.1 (Exact central-parity leakage). *If q_r, F_r, C_r denote the same-parity blocks in (5.1), then*

$$E_r = \frac{1}{2}(F_r + rC_r) - q_r^2 = ((I - P)\tilde{Q}P_r)^*((I - P)\tilde{Q}P_r). \quad (5.2)$$

Proof. Causal convolution gives $\tilde{U}^* = R_L\tilde{U}R_L$. For $R_L f = r f$, $\tilde{Q}f = -(\tilde{U}f + rR_L\tilde{U}f)/2$. Taking the full-output squared norm yields $\|\tilde{Q}f\|^2 = \langle f, (F_r + rC_r)f \rangle/2$. Subtract the retained-output norm $\|q_r f\|^2$ and polarize. $\square$

Proposition 5.2 (Certified finite test). *Suppose the exact central tail is at least aI , $a > m$, and $\|\tilde{U}P\| < K$. If both finite inequalities*

$$(a - m)(q_r - mI) - E_r - \epsilon I \succeq 0, \quad r = \pm 1, \quad (5.3)$$

hold, where

$$\epsilon = (|a - m| + 4K)\eta_M + 2\eta_M^2, \quad (5.4)$$

then $Q_{0,L} \succeq mI$ on the full form domain.

Proof. The profile difference has norm at most η_M , as does its symmetric part. On each input parity, $\tilde{Q}P_r$ is an output-parity projection of $-\tilde{U}P_r$, and hence has norm below K . The full squared-output Gram and the squared retained matrix each change by at most $2K\eta_M + \eta_M^2$. The linear head term changes by at most $|a - m|\eta_M$. Therefore (5.3) implies

$$(a - m)(Q_{P,r} - mI) - B_r^* B_r \succeq 0, \quad B_r = (I - P)Q_0 P_r,$$

for the exact central form.

For an arbitrary input in one sector, write it as $h + t$ with h retained and t in the form-domain tail. The quadratic form of $Q_0 - mI$ is bounded below by

$$\langle h, (Q_{P,r} - mI)h \rangle + 2\operatorname{Re}\langle B_r h, t \rangle + (a - m)\|t\|^2.$$

Completing the square, or minimizing over t , leaves $\langle h, (Q_{P,r} - mI - (a - m)^{-1}B_r^* B_r)h \rangle \geq 0$. The two orthogonal reflection sectors exhaust the complex Hilbert space. $\square$

The use of the exact parity leakage is consequential. Replacing it by the full leakage of $\tilde{U}$ also bounds the coupling, but charges irrelevant output parity against the available tail margin. Equation (5.2) removes that loss without discarding any coupling of the central form.

6 Finite construction and the certificate at 9/5

6.1 Polynomial and logarithmic output formulas

Use the unitary coordinate $u = x/L$, so the normalized basis is $\sqrt{2n+1} p_n(u)$, where

$$p_n(u) = \sum_{k=0}^n (-1)^{n+k} \binom{n}{k} \binom{n+k}{k} u^k. \quad (6.1)$$

With $\ell_L = \gamma + \log(2\pi L)$, the gamma derivative of a monomial is

$$U^\gamma u^k = u^k (\log u + \ell_L - H_k) + \sum_{j \geq 1} g_j L^j \frac{(j-1)! k!}{(j+k)!} u^{j+k}. \quad (6.2)$$

This follows by differentiating the beta integral for $I_\nu u^k$ and integrating each term of S_w . For the model, the sum stops at $j = M$. Every output therefore has the form

$$\tilde{U} p_n(u) = p_n(u) \log u + q_n(u) + \sum_{\log j < L} \frac{2\Lambda(j)}{\sqrt{j}} \mathbf{1}_{u > \delta_j} p_n(u - \delta_j), \quad \delta_j = \frac{\log j}{L}. \quad (6.3)$$

The polynomial q_n is supplied by (6.2). The head, F , and C are integrated using exact formulas for moments of $1, \log u, (\log u)^2, \log(1-u)$, and $\log u \log(1-u)$, including truncated integration intervals. Appendix A lists them. There is no quadrature error and no finite-output truncation.

For the reflected delayed products, the nonempty integration window is $(\delta_i, 1 - \delta_j)$, equivalently $\log(ij) < L$. At $L = 9/5$ the active generator indices are 2, 3, 4, 5, and the ordered overlap pairs are exactly (2, 2), (2, 3), (3, 2).

6.2 Enclosed arithmetic

The standalone implementation uses Arb real balls [3] through python-flint [7]. Integer and rational coefficients in (4.4), (6.1), and (6.2) are formed exactly. All subsequent conversions, elementary constants, logarithms, square roots, moments, polynomial reflections, and matrix operations are performed with balls at 1536-bit working precision. Exact symmetries are imposed only after checking consistency of the enclosures.

The matrix F encloses the model-output Gram, and its trace is checked below 10^7 . Hence $\|\tilde{U}P\| < 10000$, allowing $K = 10000$ in (5.4). The computed trace is approximately 1743.1632085; the deliberately loose bound avoids making this part of the proof delicate. An outward lower value of (4.6) and an outward upper value of (4.8) are used. An outward upper value of (5.4) is then subtracted from each model Schur matrix.

For a symmetric ball matrix M , the sign test uses the recurrence

$$d_i = M_{ii} - \sum_{k < i} \ell_{ik}^2 d_k, \\ \ell_{ji} = \frac{M_{ji} - \sum_{k < i} \ell_{jk} \ell_{ik} d_k}{d_i} \quad (j > i).$$

Every pivot must be strictly positive as a ball. All divisions then have denominators excluding zero. Induction encloses the exact LDL^* factors of the true symmetric matrix; positive pivots imply its positivity. Ordinary floating-point eigenvalues never enter this decision.

Theorem 6.1 (Computer-assisted matrix inequality). *At $L = 9/5$, $N = 128$, $M = 180$, and $m = 10^{-26}$, both inequalities (5.3) hold with $K = 10000$, the tail floor (4.6), and the error (5.4). In particular, $Q_{0,9/5} \succeq 10^{-26}I$.*

Computer-assisted proof. The supplied builder evaluates (6.3), the moment formulas in Appendix A, and the bounds of Section 4 with ball arithmetic. The following rationally rounded inequalities are checked:

$$\begin{aligned} a_{128,9/5} &> 0.69124708426711, & K_{9/5} &< 5.482684, \\ \eta_{180} &< 2.477 \times 10^{-40}, & \epsilon &< 9.908 \times 10^{-36}. \end{aligned}$$

The arithmetic norm term in (4.6) is approximately 2.39376738699. Both 64×64 ball LDL tests pass. Their minimum pivots are displayed in Table 1. Applying Proposition 5.2 gives the full-operator conclusion. $\square$

Table 1: Certificate data. Displayed pivot values are rounded summaries, not spectral lower bounds. The sign decisions use complete balls.

Quantity	Even sector	Odd sector
Retained dimension	64	64
Target central floor	10^{-26}	10^{-26}
Minimum LDL pivot	$5.6796287856 \times 10^{-5}$	$1.8170726676 \times 10^{-4}$
Every pivot strictly positive	Yes	Yes
Replay from saved balls	Pass	Pass

The largest radius among the four saved matrix enclosures was less than 9.341×10^{-215} . This finite-arithmetic uncertainty is propagated through the sign tests; it is separate from the analytic remainder. The matrix archive stores dyadic midpoint/radius pairs and checks that every reconstructed ball contains the original. Reconstructing a radius may enlarge it outward. A separate process loaded this archive and repeated both LDL tests successfully.

Proof of Theorem 1.1. Theorem 6.1 establishes (1.2) at $L = 9/5$. Zero extension into a larger interval preserves the central form: the Fourier representation and its translation-invariant kernels, including the delay correlations, do not depend on the chosen containing horizon. Thus the bound holds at every shorter horizon.

Evaluation of (3.7) at $9/5$ gives $C_{9/5} < 10.298615 < 11$. The exact rational inequality

$$10^{-26} - \frac{11}{3}(3 \times 10^{-14})^2 = 6.7 \times 10^{-27} > 5 \times 10^{-27}$$

and Proposition 3.3 prove (1.3) at the outer horizon. Causality makes every shorter transfer a compression, so it has no larger norm. Self-adjointness of $H = VR_L$ and $\|H\| = \|V\|$ give (1.4). $\square$

6.3 What the larger head resolved

The preceding 96-mode experiment at the same horizon did not pass the sufficient Schur test. To explain the improvement, Table 2 compares the scalar tail floor with the diagnostic largest eigenvalue of $q_r^{-1/2} E_r q_r^{-1/2}$. The latter measures the tail floor required by the model test at level zero. It was computed using a high-precision congruence before conversion to floating point and is a diagnostic only.

Table 2: Head/tail comparison at $L = 9/5$. Only the final column represents the complete certified sign test at $m = 10^{-26}$.

N	Tail floor	Required, even	Required, odd	Complete test
96	0.39212903	0.55845	0.54762	Not certified
128	0.69124708	0.49054476	0.47957150	Pass

Thus increasing the retained space improved both the tail floor and the remaining coupling. Failure of the earlier sufficient inequality was not evidence of a negative direction of the full form. The old matrix construction used a separate aggregate arithmetic budget restricted to at most 96 modes; that budget was not extended here. The new direct-ball calculation is independent of it. As a normalization diagnostic, all shared entries of the new head and full-output Grams agree with the earlier 96-mode point matrices to within 3×10^{-217} after conversion of the causal/Hankel signs. Those old entries are not inputs to the new positivity proof.

7 Storage, scope, and the next depth problem

Corollary 7.1 (Cumulative storage). *For any spatial split of $(0, L)$ with $L \leq 9/5$, write*

$$V_{\omega, L} = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}.$$

For $0 < \omega \leq 3 \times 10^{-14}$,

$$I - X^* X - Y^* (I - Z Z^*)^{-1} Y \succeq (1 - e^{-10^{-26} \omega}) I. \quad (7.1)$$

Proof. Let $\rho = e^{-5 \times 10^{-27} \omega}$. Theorem 1.1 gives $I - V^* V \succeq (1 - \rho^2) I$, and $\|Z\| \leq \rho < 1$. The Schur complement of the lower-right block $I - Z^* Z$ is

$$I - X^* X - Y^* Y - Y^* Z (I - Z^* Z)^{-1} Z^* Y = I - X^* X - Y^* (I - Z Z^*)^{-1} Y.$$

For a fixed first-block vector, minimizing the defect quadratic form over the second-block vector leaves a value at least $(1 - \rho^2)$ times the first-block norm squared. This proves (7.1). $\square$

In particular, the split $9/5 = \log 5 + (9/5 - \log 5)$ includes the first mixed-prime region. The corollary is obtained by certifying the whole larger operator first. It does not show that positivity of a previous slab alone forces the next storage update to be positive. That distinction separates the present finite-depth result from the intended induction problem.

7.1 A diagonal formulation of the all-depth target

Proposition 7.2. *Suppose there exist $L_j \rightarrow \infty$ and $\omega_j \downarrow 0$ such that $\|V_{\omega_j, L_j}\| \leq 1$ for every j . Then the central Weil form is nonnegative on every compactly supported smooth test function. By Weil’s criterion, such a sequence would imply RH.*

Proof. Fix a finite horizon L and a smooth test f compactly supported inside it. For all sufficiently large j , causal compression gives $\|V_{\omega_j, L} f\| \leq \|f\|$. The scalar derivative at zero on this smooth core follows by dominated convergence on a far right Laplace line: the transform of the input is rapidly decreasing, while the differentiated multiplier grows only logarithmically. It is

$$\lim_{s \downarrow 0} \frac{\|f\|^2 - \|V_{s, L} f\|^2}{2s} = Q_{0, L}[f].$$

The quotients along $s = \omega_j$ are nonnegative, so their limit is nonnegative. The horizon and compact test were arbitrary; the final implication is the classical Weil criterion [1, 2]. $\square$

The proposition requires neither a depth-independent coercivity margin nor a fixed positive shift interval. Nevertheless, constructing the entire sequence remains the missing all-depth statement; the single horizon certified here is not such a sequence.

7.2 Limitations and further calculations

The scalar tail bound pays separately for every arithmetic delay norm. This is sufficient through 9/5 once the retained space is enlarged, but does not explain the prime–gamma cancellation at unbounded depth. The small-shift interval in Theorem 1.1 is a consequence of the stated coercivity floor and a quadratic perturbation estimate; it is not asserted to be sharp. It does not connect to the previously investigated interior shift ranges of the second slab.

The next concrete depth target is total horizon $L = \log 7$, using the same full-output parity construction and adapting the retained space and profile degree as required. At that endpoint the $n = 7$ delay is zero almost everywhere; a strict increase beyond it activates the next prime. A more general result would retain structure in the tail resolvent or in the cumulative storage update instead of reducing all arithmetic tail terms to separate scalar losses.

A Moment formulas and assembly conventions

The code constructs all polynomial coefficients and integrates them against the moments below. For an integer $k \geq 0$, put $s = k + 1$, $H_s = \sum_{j=1}^s 1/j$, and $H_s^{(2)} = \sum_{j=1}^s 1/j^2$. The full-interval formulas are

$$\begin{aligned} \int_0^1 u^k du &= \frac{1}{s}, & \int_0^1 u^k \log u du &= -\frac{1}{s^2}, \\ \int_0^1 u^k (\log u)^2 du &= \frac{2}{s^3}, & \int_0^1 u^k \log(1-u) du &= -\frac{H_s}{s}, \\ \int_0^1 u^k \log u \log(1-u) du &= \frac{H_s}{s^2} - \frac{\pi^2/6 - H_s^{(2)}}{s}. \end{aligned} \tag{A.1}$$

They follow by differentiating the beta integral; the first three also follow directly by differentiating $1/s$. For $0 < \delta < 1$,

$$\begin{aligned} \int_{\delta}^1 u^k du &= \frac{1 - \delta^s}{s}, \\ \int_{\delta}^1 u^k \log u du &= -\frac{1 - \delta^s}{s^2} - \frac{\delta^s \log \delta}{s}, \\ \int_{\delta}^1 u^k \log(1 - u) du &= \frac{(1 - \delta^s) \log(1 - \delta) - H_s + \sum_{j=1}^s \delta^j / j}{s}. \end{aligned} \tag{A.2}$$

Integration by parts gives the last formula. Products of delayed polynomials use $\int_{\delta}^b u^k du = (b^s - \delta^s)/s$ on the appropriate nonempty window. For reflection, expand $(1 - u)^k$ by the exact binomial theorem.

To recover the gamma polynomial q_n in (6.3), multiply each coefficient of u^k in (6.1) by $\ell_L - H_k$, and add its contributions $g_j L^j (j - 1)! k! / (j + k)!$ at degree $k + j$. The beta factor is computed rationally with the recurrence $\beta_{1,k} = 1/(k + 1)$, $\beta_{j,k} = \beta_{j-1,k} (j - 1)/(j + k)$. The series g_j is obtained by reciprocal power-series inversion of $\sinh t/t$, multiplication by $e^{t/2}$, and subtraction of $4t \cosh(t/2)$, all over the rationals.

The unit-coordinate integrals acquire factors $\sqrt{(2i + 1)(2j + 1)}$ from normalization. The saved matrices use the causal derivative U . If instead $H_1 = UR_L$ is used, the two full-output Grams acquire the entry sign $(-1)^{i+j}$; this sign is 1 within either same-parity block. The central head and central leakage are unchanged.

B Reproducibility and review map

The accompanying review package contains the LaTeX source, bibliography, compiled manuscript, continuation guide, and the full numerical supplement. The core numerical files are:

- **certify_arb.py**: exact-rational profiles, full-output ball construction, analytic tail bounds, and ball LDL;
- **analyze_certificate.py**: rational continuation checks and diagnostic relative coupling;
- **central_matrices.json.gz**: the four saved ball matrices and analytic-bound enclosures;
- **central_certificate.json**: both complete pivot lists and the matrix-archive SHA256;
- **analysis.json**: continuation validation and separately labelled diagnostics.

The recorded numerical environment was Python 3.12.14, python-flint 0.9.0, FLINT 3.6.0, and NumPy 2.3.5. NumPy is used only for diagnostic eigenvalues. The builder uses the Python standard library and python-flint. Run from the numerical supplement directory, with assertions enabled:

```
python certify_arb.py --N 128 --M 180 --bits 1536 \
  --horizon 9/5 --floor 1e-26 --output output/rebuilt
python certify_arb.py --reuse --bits 1536 --floor 1e-26 \
  --output output/length_1p8_N128
python analyze_certificate.py output/length_1p8_N128
```

A full rebuild checks the construction as well as the final sign tests. Replay checks the signs from the saved construction enclosures. The latter does not replace review of the analytic formulas. Do not run Python with $-O$. Build times are machine-dependent; the recorded matrix construction took about 64 seconds, before archive writing and final LDL.

The largest mathematical review priorities are the normalization in (2.6) and (3.3), the common-domain energy argument, the operator-level fractional tail estimate, and the full-output parity identity. The numerical review should independently check the moment assembly, outward ball serialization, and the subtraction of (5.4) before each LDL test. No first-slab numerical archive or external compact-window certificate is required for this proof.

Acknowledgment of computational assistance

This draft and its numerical implementation were prepared with substantial assistance from OpenAI language models. The record distinguishes analytic proofs, validated arithmetic, and exploratory diagnostics. The manuscript is being circulated for mathematical and computational review.

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